Five Raters Are Not a Distribution: What Learned Judges Can Learn from Human Disagreement, and Where to Spend the Annotation Budget

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Abstract

Learned judges are increasingly trained on the full histogram of human ratings, with the stated aim of learning how people disagree. We ask what part of such a histogram a text-conditioned judge can learn, separating whether the shape beyond its location (mode and mean) is *reproducible across raters* from whether it is *predictable from the text*. On rating panels from 14 datasets, including seven software-engineering datasets, a learner-free split-half analysis finds the ordinal shape poorly reproducible: its forecasting reliability is 0.04–0.41 with 4–8 raters and below 0.3 at 40 raters (crossed item × rater upper bounds ≤ 0.39 where raters recur), below the $\frac{1}{2}$ at which the raw histogram would forecast a new rater better than its own mode and mean, and its shape-to-shape reproducibility at 40 raters is 0.06–0.22, against 0.61–0.67 on NLI. Scored against the other half's own shape, no ordinal panel shows a robustly reproducible shape beyond location: it is undetectable on several (including SummEval once the half of its items whose five crowd annotations are identical is set aside), of the size location alone produces on Dorn's readability panel, and detectable on DICES-350 only within its rater pool. There, frozen representations of three LLM families recover -1 to 1 % of it (upper bounds ≤ 12 %), although a planted text-predictable shape of the same magnitude is recovered at 15–33 %; on MNLI and SNLI they recover 28–38 % (lower bounds ≥ 24 %). Learned judges behave accordingly: replacing the shape of 69–83 % of training targets while preserving mode and mean changes LoRA judges by at most ± 0.0015 RPS (95 % bounds), whereas one-hot targets are worse on 10 of 11 panels. Beyond item location, the structure is mostly rater-side. For the annotation budget, frozen-feature judges improve with more items rather than more raters per item on most panels; in a prospective LoRA block on POPQUORN, one rating per item costs 0.006 RPS against the full data and a second rating recovers about half of it. On SummEval, whose crowd ratings are nearly uncorrelated with the held-out experts, the same contrasts measure expert dilution instead. We release per-dataset disagreement reliability cards and the estimators.

1 Introduction

Human raters disagree, and a growing line of work trains learned judges — language models fine-tuned to score text — on the full distribution of human ratings rather than on a single aggregated label (Chen et al., 2025; Singh et al., 2025; Kohli, 2026; Benito-Santos and Ghajari, 2025). The appeal is clear: a histogram records which neighbouring levels received mass and how concentrated the ratings were, and a judge that reproduces it would "capture human disagreement" rather than average it away. The data behind that appeal are thin. In most datasets the histogram is computed from three to nine people. Five ratings of a code snippet's readability are a sample from a population of possible raters, and a model trained to reproduce the sample exactly is, in part, trained to reproduce sampling noise.

This paper asks a question that the literature on learning from label variation (Uma et al., 2021; Plank, 2022) has not asked in this form: **which part of a few-rater histogram can a text-conditioned judge learn?** We decompose every histogram into its *location* (mode and mean) and the *shape beyond location* (the residual allocation of mass once mode and mean are fixed), and we measure two properties of the shape that are usually conflated:

1. **Reproducibility across raters.** If the same items were rated by a disjoint panel, would the shape come out the same? We estimate this learner-free with a split-half design: an exact identity, A1 = M − 2X,

relates the forecasting value of the raw histogram to the magnitude M of its residual relative to an exact mode-and-mean comparator and the cross-panel agreement X of that residual; the ratio $\rho = \mathrm{X/M}$ is a split-half reliability of the shape at a given number of raters, and the raw histogram beats the comparator for predicting a new rater exactly when $\rho > \frac{1}{2}$.

2. **Predictability from the text.** Even a reproducible shape is useless to a judge unless the text carries it. We fit a cross-fitted ridge decoder from a frozen LLM's representation to the panel-1 residual and score it on an independent panel-2 residual with a proper-score gain G_{own}, reported also as the fraction $G_{\text{own}}/X_{\text{own}}$ of the reproducible own-shape, against a within-cell feature-permutation control and a positive control.

Findings. (i) *The shape beyond mode and mean is poorly reproducible on ordinal scales.* Its split-half reliability is 0.04–0.41 on eight panels with 4–8 raters per item (intervals include 0 on four; on SummEval's non-duplicated items, §1 (ii), it falls to 0.10 and 0.05 with intervals including 0), and with three raters its reproducibility cannot be measured (Crupi et al. (2025)'s code-summary benchmark, MLCQ, Roy et al.): a split leaves two ratings per item, and two ratings are determined by their exact mode and mean under our tie rule, although 15–75 % of the three-rating histograms share their exact cell with a different histogram. On the dense ordinal panels it stays below 0.3 up to 40 raters (Buse–Weimer 0.22, DICES 0.25 and 0.15, Dorn's Java panel 0.29), so on no ordinal panel does the raw histogram forecast a new rater better than its own mode and mean. ChaosNLI is the contrast: the NLI shape's forecasting reliability exceeds $\frac{1}{2}$ from two to three raters and reaches 0.96–0.98 at 40. Part of this forecasting reliability is panel 1's residual predicting the other panel's location shift (§2.1); the reproducibility of the shape itself at 40 raters (ρ_own) is 0.06–0.22 on the dense ordinal panels (BW 0.13, DICES-350 0.10, DICES-990 0.06, Dorn 0.12–0.22; with rater resampling the BW and DICES-350 intervals include 0, upper bounds ≤ 0.26) and 0.61 / 0.67 on MNLI / SNLI, so the ordinal–NLI contrast holds on both measures. (ii) *No ordinal panel shows a robustly reproducible shape beyond location, and where an own-shape is detectable little of it is linearly decodable from the frozen features we tested.* A cross-fitted linear decoder on frozen features predicts panel 1's residual; we score it against panel 2's residual from *its own* mode and mean, so that the gain G_{own} measures the shape alone and not panel 2's location shift (§2.2). As a fraction of the reproducible own-shape X_{own} (item-cluster CLT intervals, §2.2; all panels and families in Table 2), it recovers: - on the ordinal panels: 0.9 % [−0.3, 2.1] on SummEval coherence on the full panel, but on its non-duplicated items the own-shape is not detectable and the ratio (0.5 % [−4.6, 5.6]) has no reproducible shape to refer to (relevance is dropped under the prespecified rule; ROUND22); 10 % [4, 16] on Dorn; and on DICES-350, whose own-shape is now detectable, −8 % [−23, 8] at 20 raters per half and 1 % [−9, 12] at 40; - on NLI: 28 % [24, 33] on MNLI and 38 % [33, 42] on SNLI.

Every NLI lower bound exceeds every ordinal upper bound (smallest margin: 24 % vs 16 %). The same holds for the gain net of the decoder's variance penalty, $(G_{\text{own}} - G_{\text{own,perm}})/X_{\text{own}}$: at most 30 % on the ordinal panels against at least 32 % on NLI at 20 raters per half. With four raters per half, ChaosNLI gives 22 % [18, 26] (MNLI) and 28 % [24, 32] (SNLI), so the NLI fractions do not depend on the larger panels; no ordinal panel offers a like-for-like partner at four raters, because none has a detectable own-shape there once SummEval's duplicated items are set aside. Qwen3-8B and OLMo-2-7B features give the same readings (SummEval full panel 1.7 % and 0.7 %; MNLI 30 % and 29 %). Taken together, the ordinal own-shape is undetectable on most panels (SummEval's non-duplicated items, POPQUORN, DICES-990), detectable on DICES-350 only within its observed rater pool, and on Dorn of the size location alone produces; where it is detectable at most 16 % of it is decoded, whereas a quarter to two-fifths of NLI's reproducible shape is. DICES-350 at 40 raters is the clean ordinal case: a planted text-predictable shape whose own-shape magnitude matches the observed one is detected in 5 of 5 draws and recovered at 15–33 %, against the observed −1 to 1 % across the three feature families (upper bounds ≤ 12 %; ROUND22 R22-5), and a location-and-threshold simulation of the panel yields no own-shape at all (mean X_{own} $-1.3{\cdot}10^{-5}$ against the observed $4.9{\cdot}10^{-5}$; ROUND22 R22-3), so here a reproducible ordinal shape beyond location exists within the observed rater pool and is not decoded.

SummEval caveat (ROUND22). In the released SummEval file, the five crowd annotations of 807 of the 1,600 items are identical on all four dimensions. The crowd mean is also essentially uncorrelated with the expert mean (r = 0.04 on coherence; the near-zero agreement is reported by Fabbri et al., 2021). On the 793 remaining items, X_{own} is −0.00013 [−0.00030, +0.00003] (coherence) and −0.00003 [−0.00016, +0.00010] (relevance). The forecasting reliability also falls: ρ is 0.35 → 0.10 [−0.13, 0.23] on coherence and 0.40 → 0.05 [−0.19, 0.25] on relevance. The full-panel SummEval reproducibility above is therefore carried by the duplicated annotations, and SummEval provides no evidence of a reproducible ordinal shape at $a = 4$. Its decoding fractions, and the plant calibrated to its magnitude, should be read in that light. Dorn's 10 % is an upper bound on decodable shape *beyond* location. At 20 raters the exact (mode, mean) cell still pools items of different latent location, and

a pure location-and-threshold simulation of Dorn's Java panel already yields a text-decodable own-shape: its $\langle \hat{r}, s_2 \rangle$ is 5.6 SE above 0, and over five such draws it averages about 95 % of the value observed on the real Java panel (ROUND21 R21-7). A planted dispersion shape whose own-shape magnitude matches the observed one is detected in only 1 of 5 draws (4 of 5 at 2.7 times that magnitude). On Dorn, then, the decodable signal is of the size that location and rater thresholds alone produce. Its 10 % neither shows decodable shape beyond location nor excludes a shape of that size, and we use it only as an upper bound. Scored as a valid forecast (projected onto its exact cell, permutation-corrected), the gain is $\leq 6{\cdot}10^{-5}$ RPS on the ordinal panels (SummEval $4{\cdot}10^{-6}$ and $8{\cdot}10^{-6}$, Dorn $3{\cdot}10^{-5}$) against 0.0065 and 0.0072 on MNLI and SNLI. This valid-forecast gain is scored against panel 2's histogram, so it also contains the cross part (text-predictable location denoising) and is not a fraction of the own-shape. On DICES-990 ($a = 20$) and POPQUORN ($a = 4$), X_{own}'s interval includes 0 and G_{own} is ≈ 0 ($\leq 5{\cdot}10^{-6}$ RPS). On HelpSteer2 and EmoBank the second half has two ratings, which are fixed by their own mode and mean, so the own-shape cannot be measured at this split. On these panels the forecasting reliability is carried by the cross-term, and no own-shape is detectable to decode at the split used. A planted text-predictable shape whose own-shape magnitude matches SummEval's is detected in 5 of 5 draws and recovered at 38–56 %; at half that magnitude it is detected in 5 of 5 (item-cluster CLT intervals; ROUND22 R22-4). On POPQUORN the same plant strength is detected in 5 of 5 draws, with recovery of 6–38 %. Our prespecified informativeness rule is stated on X. After two added plant strengths (γ 0.8, 1.0; amendments written after the first grid fell short), the smallest plant whose X reaches SummEval's observed X is detected in 5 of 5 draws, so the full-panel SummEval null would be informative; but that magnitude is carried by the duplicated crowd annotations, and on the non-duplicated items there is no own-shape to calibrate against (ROUND22); on POPQUORN the matching plant is detected in 2 of 5, so its null is not. The earlier, cross-term-inclusive ratio G/X also agrees across the three families (Table 3). (iii) *The judges we trained show no detectable use of the shape.* Replacing the shape of 69–83 % of training targets while preserving each target's exact mode and mean changes LoRA judges by −0.0001 [−0.0011, +0.0011] RPS on SummEval and +0.0000 [−0.0009, +0.0009] on POPQUORN (position-level bootstrap; user-level [−0.0014, +0.0015]), whereas destroying the location as well costs +0.012 and +0.003 (POPQUORN user-level interval [−0.0004, +0.0065]); since neither LoRA panel has a detectable own-shape, this null is expected rather than discriminating, and the discriminating case is ChaosNLI; the same intervention on ChaosNLI costs 0.005–0.017 Brier (0.016–0.052 KL). Across eleven ordinal panels and three feature families, training frozen-feature heads on the location-only comparator or on reliability-shrunk shapes changes held-out RPS by at most 0.0012 with mixed signs; other smoothing constructions move it by −0.005 to +0.009 in panel-dependent directions; one-hot targets are worse on 10 of 11 panels. A user-held-out replication on POPQUORN, where no evaluator contributes to any target, gives the same signs. (iv) *Beyond location, rating structure is mostly rater-side.* On the four panels with persistent rater identities, rater-specific cut-points improve held-out likelihood by 0.05–0.16 nats per rating while item-specific dispersion adds −0.017 to +0.012. Whether the residual shape itself is rater-side is panel-dependent: simulations at 40 raters show thresholds reproduce it where location alone does not on DICES-350, location alone suffices on Buse–Weimer, and item dispersion is needed on Dorn's Java panel. (v) *Budget.* At a fixed number of ratings, frozen-feature heads improve with more items rather than more raters per item on four of the five ordinal panels with more than 900 items and on both ChaosNLI panels (SummEval is flat; panels with $\leq$ 210 training items reverse). On SummEval, prospective blocks with independent rating assignments and a declared 0.005-RPS margin find LoRA judges trained on one rating per item worse than those trained on the full seven-rating data, by +0.0064 [+0.0045, +0.0085] on Qwen2.5 and +0.0066 [+0.0044, +0.0089] on OLMo-2, so equivalence and non-inferiority both fail on both bases. Because SummEval's crowd ratings are nearly uncorrelated with the held-out experts, these SummEval contrasts measure how far the expert ratings are diluted rather than the value of a rating as such (§5). Earlier blocks with a shared assignment could not distinguish them. At a matched budget, 960 × 1 beats 144 × 7 on Qwen2.5 (+0.010 [+0.005, +0.016]; shared-assignment blocks 0.023 and, on OLMo-2, 0.0095), but not significantly on OLMo-2 in the prospective block (+0.006 [−0.001, +0.013]); on POPQUORN (same-batch re-run, and again in a prospective block with independent assignments) a single rating per item is worse than the full data (+0.0063 [+0.0030, +0.0099] prospectively), a second rating recovers about half of that gap (49–62 %), and items-first is not supported. A prediction we registered — that multi-rater items would pay off at fixed budget where the shape is text-predictable — was falsified on ChaosNLI; we therefore do not derive the budget result from the mechanism and report the task-dependence of the marginal value of extra ratings as an observation.

Contributions. (1) Two reliabilities for disagreement — across raters and across text — with learner-free, cross-fitted estimators. (2) Disagreement reliability cards for every panel, telling a dataset builder whether the

shape of disagreement is reproducible at the number of raters they can afford. (3) A rater–item decomposition of the ordinal structure beyond location. (4) Fixed-budget evidence: for frozen-feature heads in three model families, more items beat more raters per item on four of the five ordinal panels with more than 900 items (SummEval is flat) and on both NLI panels. For end-to-end LoRA judges on POPQUORN, whose rating positions are homogeneous, one rating per item costs 0.006 RPS against the full data in a prospective block, a second rating recovers about half of it, and items-first is not supported; on SummEval the same contrasts measure how far expert ratings are diluted by nearly uninformative crowd ratings (§5) and are not budget evidence. The donor-replacement audit (§3) is the evidence that current training does not use the shape; shrinkage of the histogram survives as a tool for forecasting a new rater, not as a training method.

What we do not claim. That disagreement is noise (part of it is reproducible); that the text contains no residual information (we test linear decodability from three frozen model families); a theoretical rater-count threshold (a two-term 1/a model was falsified); a causal role of rater thresholds or of label structure in the ordinal–nominal difference; or a universal budget rule (Zhang et al. (2021) find the opposite when a large single-label base already exists; see §5).

2 b. Related work

Learning from label variation. Human label variation is increasingly treated as signal rather than noise (Aroyo & Welty 2015; Pavlick & Kwiatkowski 2019; Uma et al., 2021; Plank, 2022), and soft-label training is reported to improve calibration and robustness (Peterson et al., 2019; Collins et al., 2022; Singh et al., 2025). For learned judges, Chen et al. (2025) train on human rating distributions; Kohli (2026) shows that on ChaosNLI the item-specific information in soft labels saturates around ten annotators for KL. These studies compare training targets; none separates the part of the histogram that is reproducible across raters from the part that is predictable from the input, which is the distinction our results turn on.

Predicting disagreement. Disagreement is partly predictable from text: NLI label distributions can be predicted to a degree (Zhou et al., 2022), reward models detect diverging preferences with AUROC around 0.6 (Zhang et al., 2025), and demographic information helps predict who disagrees (Wan et al., 2023). Our decoding analysis gives the matching quantity for the shape beyond mode and mean, normalised by its reproducible part, and finds the NLI and ordinal cases on opposite sides.

Reliability of aggregates and rater effects. Reliability of aggregated annotations depends on the number of raters (Wong and Paritosh, 2022); rater severity and scale use are classical facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011). We use the split-half logic and the rater facet, but apply them to the shape beyond location and connect them to what a text-conditioned judge can learn.

Annotation budgets. Whether to label more items or relabel existing ones is an old question in crowdsourcing (Sheng et al., 2008; Lin et al., 2014); for hard-label classifiers, labelling many items once is preferable above a worker-quality threshold (Khetan et al., 2018). In NLP, Zhang et al. (2021) find the opposite on ChaosNLI: replacing a few thousand single-label items by a few hundred ten-label items reduces KL — in a regime where a large single-label base already exists and the evaluation targets the full distribution. Our budget results concern the complementary regime (every rating competes for items, ordinal quality ratings, judges trained only on the budgeted data), and on ChaosNLI we also observe a larger marginal value of a second rating than on ordinal panels; the two findings are consistent once the regime is stated.

3 Two reliabilities of a disagreement histogram

Setting and notation. Item i has ratings on K ordered levels from a panel of raters; the empirical histogram is h_i. For ordinal scales we score forecasts with the ranked probability score, $S(f, y) = \sum_k (F_f(k) - \mathbf{1}[y \leq k])^2/(K-1)$, and write C for the cumulative operator; for nominal tasks (ChaosNLI) S is the Brier score and C the identity. The **location** of a histogram is its exact mode and mean (nominal: modal class and modal mass); the **shape beyond location** is what remains once these are fixed. To isolate it we use the *exact-T comparator* q_T (Appendix A): a cross-fitted forecast built from other items in the same exact mode-and-mean cell (KL-projected onto the cell when the cell is sparse), so that q_T has the same mode and mean as h_i and differs from it only in shape. The residual of a histogram is $r = C(h - q_T)$. Appendix B (generated from the result directories) lists every panel with its scale, number of items and rating columns, and the analyses it enters;

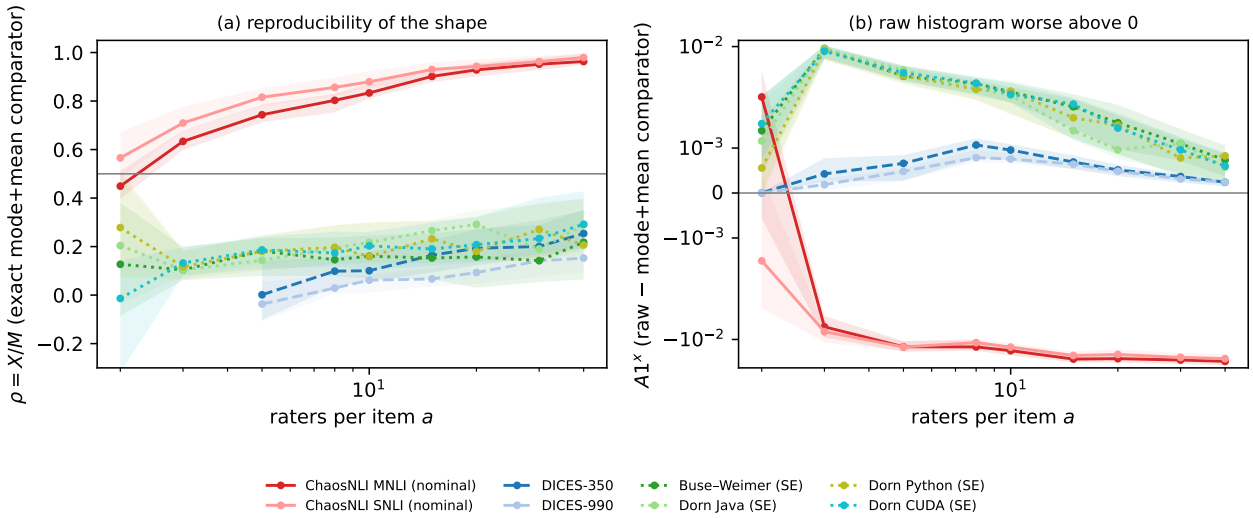


Figure 1: Reliability of the shape beyond exact mode and mean against the number of raters per item, dense panels. (a) Split-half reliability $\rho = X/M$ relative to the exact mode-and-mean comparator, with 95% item-cluster intervals; omitted where the residual is structurally empty or the ratio interval is wider than 1 (DICES with 2–5 ratings on a 3-level scale). The raw histogram beats the location-only forecast only above $\frac{1}{2}$. (b) $A1^x$: positive means the raw histogram forecasts a new rater worse than its own mode and mean.

panel counts differ across analyses because each analysis has its own eligibility (persistent rater identities, more than 900 items, an informative X).

2.1 Reproducibility across raters (split-half reliability).

In each of R repetitions the ratings of every item are split at random into disjoint panels P_1 (size a) and P_2. Writing $r_1 = C(h_1 - q_T)$ and $r_2 = C(h_2 - q_T)$, with q_T computed from P_1, the forecasting value of the raw histogram over the comparator obeys the exact identity

$$A1 \;=\; \mathbb{E}\big[S(h_1, y) - S(q_T, y)\big]_{y \in P_2} \;=\; M - 2X, \qquad M = \|r_1\|^2/n,\; X = \langle r_1, r_2\rangle/n,$$

where $\langle u, v\rangle = \sum_k u_k v_k/(K-1)$ for ordinal scales (the RPS normalisation; 1 for nominal), so the identity is exact as implemented. Thus $\rho = X/M$ is a **new-rater forecasting reliability**: the raw histogram beats the location-only forecast exactly when $\rho > \frac{1}{2}$. Because r_2 is taken against panel 1's comparator, X is not purely shape-to-shape agreement: writing q_j for the comparator built from panel j, $X = X_{\text{own}} + X_{\text{cross}}$ with $X_{\text{own}} = \langle r_1, C(h_2 - q_2)\rangle/n$ (agreement of the two panels' shapes, each beyond its own location) and $X_{\text{cross}} = \langle r_1, C(q_2 - q_1)\rangle/n$ (panel 1's residual predicting the other panel's location shift). On the dense panels we report both (Appendix A); the cross-term is not negligible (32–40 % of X on ChaosNLI at $a = 20$; most of X on the ordinal dense panels at $a \le 8$), so we call ρ a forecasting quantity and use $\rho_{\text{own}} = X_{\text{own}}/M$ for the reproducibility of the shape itself. On the small panels the second panel is a single rating, its own comparator is that rating, and $X_{\text{own}} \equiv 0$: there ρ is by construction a statement about forecasting a new rater. ρ is specific to the number of raters a and, as a ratio, unstable when M is tiny; we report it with intervals and omit it where the residual is structurally empty (e.g. two or three ratings on a three-level scale). We also report the support of the donor-replacement intervention (the fraction of items whose histogram could be changed at all while preserving mode and mean). A smoother, response-based comparator that does not reproduce the noisy location gives larger values of ρ because it also credits location reliability; we use the exact comparator throughout. Intervals are item-cluster bootstraps in which all copies of a resampled item stay in one cross-fitting fold and the comparator is refitted in every replicate. *Population.* On panels whose rating columns are positions rather than persistent people (HelpSteer2, EmoBank, CSC, ChaosNLI, Dorn and the SE panels without identities), the item bootstrap resamples an item's ratings with the item, so the intervals refer to new raters *if* the anonymous positions are filled by different people independently across items — an assumption the datasets do not document. SummEval is mixed: its three expert columns are persistent people (intervals condition on them), its five crowd columns

Table 1: Disagreement reliability cards (exact mode-and-mean comparator). $\rho = X/M$: split-half new-rater forecasting reliability of the shape beyond mode and mean at a raters, 95% item-cluster interval. $\rho_{\text{own}} = X_{\text{own}}/M$ (dense panels, $a = 40$ vs 20 raters): agreement of the two panels' shapes, each beyond its own mode and mean (§2.1); on small panels the held-out panel is one rating and $X_{\text{own}} \equiv 0$. $A1^x = M - 2X$: score change of the raw histogram over the exact comparator when forecasting a held-out rating (negative = raw histogram better). The raw histogram forecasts a new rater better than its own mode and mean only if $\rho > \frac{1}{2}$ ($A1^x < 0$). Support: share of items whose histogram can be changed at all with mode and mean fixed. Three-rater panels: the mode-and-mean cell determines the histogram (support 0). SummEval: on 807 of the 1,600 released items the five crowd annotations are identical on all four dimensions; the non-dup. rows use the other 793 items (§1). SE = software engineering.

panel	raters	a	ρ [95% CI]	ρ_{own} [95% CI]	$A1^x$	support
CSC sarcasm	6	5	0.04 [-0.06, 0.13]	–	+0.0050	0.72
EmoBank	5	4	0.41 [0.27, 0.55]	–	+0.0003	0.14
HelpSteer2	5	4	0.13 [0.04, 0.17]	–	+0.0036	0.45
POPQUORN off.	8	7	0.07 [-0.04, 0.25]	–	+0.0031	0.69
POPQUORN pol.	5	4	0.04 [-0.03, 0.12]	–	+0.0044	0.42
Scalabrino^{SE}	9	8	0.19 [-0.24, 0.55]	–	+0.0027	0.69
SummEval coh.	8	7	0.35 [0.24, 0.48]	–	+0.0016	0.87
SummEval coh. (non-dup.)	8	7	0.10 [-0.13, 0.23]	–	+0.0032	–
SummEval rel.	8	7	0.40 [0.30, 0.54]	–	+0.0008	0.79
SummEval rel. (non-dup.)	8	7	0.05 [-0.19, 0.25]	–	+0.0031	–
Buse–Weimer^{SE}	121	40	0.22 [0.06, 0.35]	0.13 [0.04, 0.24]	+0.0007	0.89
DICES-350	104	40	0.25 [0.19, 0.35]	0.10 [0.02, 0.19]	+0.0002	1.00
DICES-990	60	40	0.15 [0.10, 0.24]	0.06 [-0.00, 0.14]	+0.0002	1.00
Dorn CUDA^{SE}	160	40	0.29 [0.20, 0.43]	0.22 [0.11, 0.29]	+0.0006	–
Dorn Java^{SE}	160	40	0.29 [0.16, 0.40]	0.12 [0.07, 0.26]	+0.0006	–
Dorn Python^{SE}	160	40	0.21 [0.18, 0.40]	0.13 [0.09, 0.25]	+0.0008	–
ChaosNLI MNLI	100	40	0.96 [0.95, 0.99]	0.61 [0.58, 0.63]	−0.0186	1.00
ChaosNLI SNLI	100	40	0.98 [0.95, 1.00]	0.67 [0.64, 0.69]	−0.0173	0.99
Crupi CA^{SE}	3	2	– (no shape)	–	–	0.00
Crupi conc.^{SE}	3	2	– (no shape)	–	–	0.00
Crupi flu.^{SE}	3	2	– (no shape)	–	–	0.00
MLCQ^{SE}	3	2	– (no shape)	–	–	0.00
Roy CA^{SE}	3	2	– (no shape)	–	–	0.00

are positions; the crowd mean is nearly uncorrelated with the expert mean (r = 0.04; Fabbri et al., 2021), and on half of the released items the five crowd annotations are identical on all four dimensions, an undocumented pattern we could not verify upstream (§1, ROUND22). Where raters recur across items (Buse–Weimer, DICES-350, DICES-990, Scalabrino, POPQUORN), item-cluster intervals condition on the observed rater pool; for the three complete crossed panels we also report crossed item × rater bootstrap intervals (Appendix A): the ρ upper bounds stay below the $\frac{1}{2}$ forecasting threshold (BW 0.39, DICES-350 0.36 at 40 raters; DICES-990 0.10 at 10 raters, the largest feasible; rounded outward) — this supports "below $\frac{1}{2}$" for new raters, not the "below 0.3" of the point estimates, but ρ_own's intervals then include 0 on BW and DICES-350 ([−0.004, 0.26], [−0.02, 0.19]), so for new raters we bound the ordinal shape-to-shape reproducibility above rather than show it positive, and for POPQUORN the LoRA comparisons are additionally given under user-level clustering (§5). Table 1 gives the resulting *reliability cards*; Figure 1 traces ρ and $A1^x = M - 2X$ (the score change of the raw histogram over the exact comparator when forecasting a held-out rating; negative means the raw histogram is better) against a on the dense panels.

2.2 Predictability from the text (decodability).

A judge sees the text, not the raters. Whether the reproducible shape is available to it is a separate question, which we answer with the representation the judges are built on: last-token hidden states of the frozen Qwen2.5-7B-Instruct (Qwen2.5-Coder for code panels) under the same rating prompt. A multi-output ridge decoder, group-cross-fitted over five item folds with a label-free PCA and a nested penalty, predicts r_1 from the features; it is scored on the *independent* panel r_2 by the covariance $V = \langle \hat{r}, r_2 \rangle / n$ and by the proper-score gain

$$G \;=\; \big(2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2\big)/n \;=\; \mathbb{E}_{y \in P_2}\big[S(q_T, y) - S(q_T + C^{-1}\hat{r},\, y)\big],$$

the exact reduction in score obtained by adding the decoded shape to the location-only forecast. Two controls accompany every estimate: a *within-cell permutation* of the features among items with the same exact mode and mean (which must drive G to zero or below), and a *positive control* that predicts the panel-1 mean with the same pipeline (which must be clearly positive, else the representation or the alignment is broken). A third control calibrates power: we plant a known, text-predictable shape (item dispersion log-linear in a random direction of the features, with the panel's fitted locations and cut-points) and rerun the unchanged pipeline. **Own-shape decomposition (ROUND20 R20-1).** With $s_2 = C(h_2 - q_2)$ (panel 2's residual from its own exact cell) and $d = C(q_2 - q_1)$, the gain splits as $G = G_{\text{own}} + G_{\text{cross}}$, with $G_{\text{own}} = (2\langle \hat{r}, s_2 \rangle - \|\hat{r}\|^2)/n$ and $G_{\text{cross}} = 2\langle \hat{r}, d \rangle / n$, and likewise $X = X_{\text{own}} + X_{\text{cross}}$ (§2.1). The headline estimand is $G_{\text{own}}/X_{\text{own}}$, reported where X_{own}'s interval excludes 0; we call G/X the *normalised forecasting gain*. Reading the planted control on this scale: with no planted shape the permutation-corrected projected gain ΔGP is zero on both panels in every draw. G_{own} is slightly negative, as it must be: its expectation under the null is $-E\|\hat{r}\|^2/n$, the decoder's variance penalty. Our frozen null check, which required its interval to cover 0, was therefore ill-posed; the uncorrected projected gain GP is positive under the null (the smoothing artefact of §2.2). Over five simulation draws on SummEval (ROUND20 R20-2), the planted shape is detected (item-cluster CLT lower bound of G_own > 0; ROUND22 R22-4) in 0 of 5 at $\gamma = 0$, 5 of 5 at $\gamma = 0.3$ and 5 of 5 at $\gamma = 0.6$. At $\gamma = 0.6$, X_{own} is 0.00033–0.00047, bracketing SummEval's observed 0.00038, and the decoder recovers 38–56 % of the planted own-shape (POPQUORN, five draws: detected in 5 of 5, 6–38 %). The rule we fixed in advance compares the planted X with the observed X. The first grid fell short (strongest plant 0.00166 against 0.00198), so we added $\gamma = 0.8$ and 1.0 (ROUND20 R20-2a/c, written after seeing that). At $\gamma = 1.0$ (mean planted X 0.00225) the plant is detected in 5 of 5 draws and recovered at 31–57 %, so at its full-panel magnitude the SummEval null would be informative under the rule; but that magnitude is carried by the duplicated crowd annotations (ROUND22), so the calibration has no reproducible SummEval shape to refer to. On POPQUORN the reference plant ($\gamma = 0.3$, X 0.00060 against the observed 0.00036) is detected in 2 of 5, so its null is uninformative. The planted own-shape saturates near 0.0004–0.0005 for $\gamma \geq 0.6$, close to SummEval's observed 0.00038; the 7–12 % recovery under G/X came from the cross-term in the denominator. Two features of the estimator are post hoc and disclosed. First, at $\gamma = 0$, X_{own} is about $-1{\cdot}10^{-4}$ although the halves are exchangeable ($a = b = 4$). Simulations from the fitted rater–item model locate the cause (ROUND21 R21-2). Each item's two halves are drawn without replacement from one fixed pool of raters with different cut-points, so the halves' rater mixes are anti-correlated. With identical cut-points, or with raters drawn afresh for each rating, the bias vanishes (SummEval -0.000099 ± 0.000010 vs $+0.000007$ and $+0.000018$; DICES-350 -0.000015 ± 0.000003 vs ≈ 0; none on the panels fitted with one rater group). The real X_{own} is therefore reproducibility within the observed rater pool, biased downward for new raters. That makes "no detectable own-shape" easier to declare: corrected for the pool effect, SummEval's full-panel X_{own} is about 0.00048 (a correction from a rater model that cannot produce the duplicated crowd annotations, so it does not describe them), and DICES-350's interval at $a = 20$ only just covers 0. Second, the item bootstrap of G_{own} duplicates items inside the cross-fit and is not centred on its point estimate. We therefore use item-cluster CLT intervals on the per-item contributions, averaged over ten cross-fitting repetitions with the cross-fitted predictions held fixed, so that no item is ever duplicated (ROUND21 R21-1). Ratios get delta-method intervals. As a check, fixed in advance, the interval for $\langle \hat{r}, s_2 \rangle / n$, which has mean 0 under the no-plant null, covers 0 in 9 of 10 null draws. A nonlinear decoder (approximate Gaussian-kernel ridge, 2,048 random Fourier features on the same PCA scores; ROUND19 R19-5) does not change this. On SummEval and POPQUORN its null is calibrated. Its planted-shape recovery is no better than ridge: the lower bound at the strongest plant stays below the ridge point estimate. On the real panels its G/X upper bounds are 0.2 % and 0.4 %, as for ridge. The decoded forecast $q_T + C^{-1}\hat{r}$ need not be a distribution in q_T's cell. It has negative mass on 3–82 % of items, a changed weak mode on ≤ 6 % of ordinal items, and on nominal panels a changed modal mass. We therefore also project it onto the cell: nearest point in the RPS metric with $q \geq 0$, $\sum q = 1$, q_T's mean and weak mode, or its modal mass for nominal panels. The projection alone gains score with no text information, because it smooths toward the interior. We correct for this by subtracting the projected gain of the within-cell-permuted forecast, which is zero under the planted null. This permutation-corrected projected gain is reported beside G (ROUND19 R19-8; point estimates). Directions are symmetrised ($P_1 \leftrightarrow P_2$) and bootstrapped over item clusters. Because a single layer is a weak probe, we repeat the analysis with per-layer mean- and last-token features and greedy selection of the best layers (the probing recipe of Li et al., 2026).

2.3 Why the two can come apart.

ρ asks whether the shape is a property of the item's *rater population*; G asks whether it is a property of the item's *text*. A shape can be reproducible yet invisible to text — for instance when it is produced by how particular kinds of raters use the scale — and a text-predictable shape can be unreliable with few raters. Section 4 examines how far a rater-side account explains the first case on the ordinal panels. Nothing in §2.1 or §2.2 trains a judge; both are measurements of the data and the representation, and both are released with the reliability cards so that a dataset builder can compute them before deciding how many raters to pay for.

4 What learned judges use: the audit, condensed

If a judge learned the shape beyond location, destroying that shape in its training targets would hurt it. A donor-replacement audit (Appendix A gives the full design and tables) does exactly this with certified dose: for 69–83 % of training items the histogram is replaced by another item's histogram projected onto the recipient's exact mode and mean (mean TV 0.30–0.35 among changed items), under two learners. **End-to-end LoRA judges** (Qwen2.5-7B-Instruct, the distributional objective of Chen et al., 2025; three optimizer–assignment realisations × three held-out-rater folds) change by −0.0001 [−0.0011, +0.0011] RPS on SummEval coherence and +0.0000 [−0.0009, +0.0009] on POPQUORN offensiveness, whereas destroying the location as well (mode-only replacement) costs +0.0120 [+0.0086, +0.0155] and +0.0029 [+0.0009, +0.0050] with every seed-by-fold cell positive (POPQUORN intervals resample rating positions; resampling users instead widens them to [−0.0014, +0.0015] and [−0.0004, +0.0065]). **Frozen-feature heads** on ten ordinal panels move by −0.0024 to +0.0006. On SummEval (once the duplicated crowd annotations are set aside) and on POPQUORN no own-shape is detectable at all (§2.2), so there a null is the expected outcome rather than independent evidence that judges ignore a learnable shape; the discriminating case is ChaosNLI, where the shape is reproducible and decodable. On the nominal ChaosNLI panels the same kind of intervention — scrambling the allocation of mass beyond the modal class while preserving the modal mass — raises the test Brier score (the paper's nominal score) by 0.005–0.012 on MNLI and 0.008–0.017 on SNLI at every annotation count (KL 0.016–0.052), i.e. by 0.29–1.31 of the Brier gap between an oracle that uses the item's own training histogram and the exact comparator on the same test items (ROUND19 R19-7); on binary αNLI, where the replacement cannot change any target, the same contrast is a sham and averages −0.003 to +0.002 (per-configuration SD up to 0.014, since optimisation noise is not matched across arms), so only the nine-configuration averages are interpreted.

A stronger version of the same test asks whether *any* construction of the per-item target helps. We trained the heads on eleven ordinal panels with the raw histogram, the location-only comparator, reliability-shrunk shapes and locations, Dirichlet smoothing toward the mode prototype (the mean histogram of other-fold items with the same mode), label smoothing toward uniform and toward the marginal, and a cross-item smoothed comparator, thirty configurations each (5 seeds × 3 group-disjoint item splits × 2 annotation-position splits). The location-only comparator and reliability-shrunk shapes change held-out RPS by at most 0.0012 with mixed signs; the other smoothing constructions move it by −0.005 to +0.009 with signs that differ across panels; the one-hot target is worse on 10 of 11 panels (up to +0.010; Scalabrino −0.0013). The learner pools location across items on its own and shows no sensitivity to the per-item shape — what §2.2 leads one to expect for a shape that is not linearly decodable from its features. In these audits the ordinal-scale judges show no detectable sensitivity to the per-item shape beyond location, whereas on NLI the same intervention is costly, consistent with the decoding result there.

5 Where the ordinal shape comes from: a rater–item decomposition

The split-half analysis says the few-rater shape is mostly noise for forecasting a new rater ($\rho > 0$ with an interval excluding 0 on four of the eight small panels, and on two once SummEval's duplicated items are set aside), and on the dense ordinal panels the shape-to-shape reproducibility at 40 raters is small (ρ_own 0.06–0.22 with item-only intervals; with rater resampling the BW and DICES-350 intervals include 0); the decoding analysis says little of it is in the text. What is it, then? We fit three ordered-probit models to the rating matrices of the panels with persistent rater identities (Buse–Weimer, DICES-350, Scalabrino, the SummEval experts) and location-plus-dispersion models to the position panels: $Y^*_{ir} = \theta_i + \sigma_i\,\varepsilon_{ir}$ with cut-points c_r; **M0** item location with common cut-points, **M1** adding rater-specific cut-points (harshness), **M2** adding item-specific dispersion

Table 2: Own-shape decoding: the cross-fitted decoder's gain against panel 2's residual from its own mode and mean, as a percentage of the reproducible own-shape, $100\,G_{\text{own}}/X_{\text{own}}$, and net of the within-cell permutation (decoder variance penalty); X_{own} in units of 10^{-4}. Item-cluster CLT 95% intervals (§2.2). a/b: raters in panel 1 / panel 2. No ratio where X_{own}'s interval covers 0. SummEval full-panel rows include the duplicated crowd annotations; non-dup. rows exclude them (§1). HelpSteer2 and EmoBank ($b = 2$) cannot measure the own-shape.

panel	a/b	features	X_{own}	$G_{\text{own}}/X_{\text{own}}$ (%)	net (%)
DICES-350	20/20	Qwen2.5	+0.44 [+0.05, +0.83]	−8.0 [−23.0, +7.1]	+11.6 [−6.2, +29.4]
DICES-350	20/20	OLMo-2	+0.44 [+0.05, +0.83]	−4.7 [−13.2, +3.7]	+8.9 [−2.7, +20.5]
DICES-350	20/20	Qwen3	+0.44 [+0.05, +0.83]	−0.4 [−7.5, +6.8]	+8.8 [−1.9, +19.4]
DICES-350	40/20	Qwen2.5	+0.49 [+0.16, +0.82]	+1.3 [−8.8, +11.4]	+15.0 [+2.6, +27.4]
DICES-350	40/20	OLMo-2	+0.49 [+0.16, +0.82]	−0.0 [−7.2, +7.2]	+6.8 [−1.9, +15.6]
DICES-350	40/20	Qwen3	+0.49 [+0.16, +0.82]	−0.6 [−8.0, +6.8]	+9.5 [−0.3, +19.3]
DICES-990	20/20	Qwen2.5	+0.22 [−0.02, +0.45]	X_{own} n.s.	–
Dorn (all)^{SE}	20/20	Qwen2.5	+2.73 [+2.03, +3.43]	+10.2 [+4.4, +15.9]	+11.4 [+7.1, +15.8]
Dorn Java^{SE}	20/20	Qwen2.5	+2.69 [+1.55, +3.84]	+0.3 [−14.9, +15.5]	+11.3 [+2.7, +19.9]
EmoBank	3/2	Qwen2.5	+0.02 [−0.02, +0.06]	X_{own} n.s.	–
HelpSteer2	3/2	Qwen2.5	+0.00 [+0.00, +0.00]	X_{own} n.s.	–
POPQUORN off.	4/4	Qwen2.5	−0.08 [−1.14, +0.99]	X_{own} n.s.	–
SummEval coh. (full panel)	4/4	Qwen2.5	+3.75 [+2.45, +5.05]	+0.9 [−0.3, +2.0]	+2.2 [+1.0, +3.4]
SummEval coh. (full panel)	4/4	OLMo-2	+3.75 [+2.45, +5.05]	+0.7 [−0.4, +1.8]	+3.1 [+1.5, +4.7]
SummEval coh. (full panel)	4/4	Qwen3	+3.75 [+2.45, +5.05]	+1.7 [+0.6, +2.8]	+3.1 [+1.6, +4.6]
SummEval coh. (non-dup.)	4/4	Qwen2.5	−1.33 [−2.98, +0.31]	X_{own} n.s.	–
SummEval rel. (full panel)	4/4	Qwen2.5	+3.07 [+1.83, +4.31]	+1.3 [+0.1, +2.5]	+2.9 [+1.4, +4.4]
SummEval rel. (non-dup.)	4/4	Qwen2.5	−0.32 [−1.64, +1.00]	X_{own} n.s.	–
ChaosNLI MNLI	4/4	Qwen2.5	+41.62 [+36.40, +46.84]	+22.0 [+18.7, +25.4]	+30.7 [+26.8, +34.7]
ChaosNLI MNLI	20/20	Qwen2.5	+111.31 [+102.32, +120.31]	+28.3 [+24.6, +32.0]	+36.5 [+32.8, +40.2]
ChaosNLI MNLI	20/20	OLMo-2	+111.31 [+102.32, +120.31]	+28.9 [+25.4, +32.4]	+36.4 [+32.9, +39.9]
ChaosNLI MNLI	20/20	Qwen3	+111.31 [+102.32, +120.31]	+30.2 [+26.7, +33.8]	+38.2 [+34.7, +41.8]
ChaosNLI SNLI	4/4	Qwen2.5	+35.23 [+30.41, +40.05]	+28.4 [+24.8, +31.9]	+36.4 [+32.1, +40.6]
ChaosNLI SNLI	20/20	Qwen2.5	+111.94 [+102.04, +121.84]	+37.8 [+34.0, +41.6]	+45.2 [+41.2, +49.1]

σ_i (the inverse of IRT discrimination). Held-out log-likelihood over random cells, five folds, with the dispersion penalty chosen on held-out data.

Rater thresholds carry the structure: M1 improves held-out likelihood by +0.111 (BW), +0.146 (DICES-350), +0.156 (Scalabrino), +0.104 / +0.047 (SummEval coherence / relevance) nats per rating; item dispersion adds −0.017 (Scalabrino) to +0.012 and over-fits on the 5–7-rating position panels. Simulating rating matrices from the fitted models and re-running the split-half analysis at the dense splits (40 + 20 raters, 20 simulations per model; ROUND19 R19-3) gives a panel-dependent answer. On DICES-350 the observed X_{extown} lies within the simulation range of the threshold model M1 but not of the location-only model M0, so thresholds explain something location alone cannot. On Buse–Weimer the location-only model already reproduces it, so thresholds are not needed for the residual even though they carry the likelihood gain. On Dorn's Java panel (no rater identities) only the dispersion model M2 reproduces it. At smaller splits ($a = 10$) the location-only model reproduces most of the observed X through $X_{extcross}$ (BW 0.00037 of 0.00052; Dorn Java 0.00056 of 0.00068): much of the small-a positive X is location noise interacting with the exact comparator, not shape. SummEval retains a small residual that neither thresholds nor one-dimensional dispersion reproduce; it is carried by the items whose five crowd annotations are identical, which no rater model generates, and it disappears on the remaining items (§1, ROUND22). The mechanism is the textbook one of many-facet Rasch measurement (Linacre, 1989; Eckes, 2011), and we present it as measurement, not as a claim about raters' psychology; for judges it means that, on the panels with persistent rater identities, most of the held-out likelihood structure beyond item location is associated with who rated; whether the residual shape itself is rater-side is established only on DICES-350, and only conditional on its observed rater pool (with rater resampling its ρ_{own} interval includes 0, §2.1).

The annotation side confirms the same picture from a different angle. For forecasting a new rater from an item's own ratings, shrinking both location and shape toward the mode prototype by their split-half reliabilities improves RPS by −0.0048 [−0.0052, −0.0045] over the raw histogram (eight panels, inverse-variance pooled; descriptive only — the panels share documents and annotators and the pooled interval ignores that covariance), a cross-item smoothed comparator by −0.0117 [−0.0123, −0.0112]; shrinking the *shape* alone by its reliability

Table 3: Decodability of the residual beyond exact mode and mean from frozen representations of three model families: decoded proper-score gain on an independent rater panel as a percentage of the panel's reproducible shape, $100\,G/X$ (95% item-cluster bootstrap); positive control in parentheses (Spearman of the decoded vs held-out mean; point estimate). "X n.s.": the reproducible shape itself has an interval covering 0, so no ratio is reported. Percentile intervals and the cross-term-inclusive G/X are kept for comparison; the own-shape fractions with CLT intervals are in Table 2. SummEval rows include the duplicated crowd annotations.

panel	Qwen2.5-7B	Qwen3-8B	OLMo-2-7B
Buse–Weimer^{SE}	+6.6 [-7.0, +26.0] (0.54)	–	–
CSC sarcasm	X n.s. (0.44)	X n.s. (0.51)	X n.s. (0.52)
DICES-350	-2.4 [-18.5, +1.4] (0.47)	–	–
DICES-990	-0.1 [-24.6, +3.2] (0.71)	-2.9 [-13.8, +4.6] (0.72)	-3.3 [-7.9, +2.9] (0.75)
Dorn (all)^{SE}	+3.2 [-2.7, +5.4] (0.67)	–	–
EmoBank	+0.9 [-0.8, +2.0] (0.48)	+0.6 [-0.8, +1.8] (0.48)	+1.1 [-1.0, +1.7] (0.49)
HelpSteer2	+2.5 [-5.0, +10.2] (0.36)	+4.1 [-3.9, +9.8] (0.39)	+0.9 [-7.6, +7.3] (0.27)
JetBrains naming^{SE}	+1.1 [-37.3, +2.9] (0.30)	–	–
POPQUORN off.	-1.6 [-4.9, +0.0] (0.65)	-0.8 [-4.2, +0.4] (0.65)	-2.0 [-4.9, +0.1] (0.68)
POPQUORN pol.	X n.s. (0.73)	–	–
Scalabrino^{SE}	X n.s. (0.41)	–	–
SummEval coh.	-0.1 [-1.0, +0.2] (0.41)	+0.1 [-0.9, +0.3] (0.36)	-0.2 [-1.5, +0.2] (0.27)
SummEval rel.	-0.1 [-2.5, +0.4] (0.33)	–	–
ChaosNLI MNLI	+16.9 [+11.2, +17.6] (0.29)	+18.2 [+13.3, +19.1] (0.32)	+17.7 [+12.4, +18.3] (0.24)
ChaosNLI SNLI	+26.0 [+20.4, +27.0] (0.41)	–	–

Table 4: Frozen-feature heads at a fixed rating budget (Qwen2.5-7B features; soft-label head): test RPS (ordinal; ChaosNLI: KL to a held-out 50-rater reference) for items × ratings per item at the same total, mean over 30 configurations (5 seeds × 3 item splits × 2 rater splits). Lower is better; best per row in bold.

panel	ratings	items × ratings: score
Buse–Weimer^{SE}	60	60×1: 0.187, **30×2: 0.171**
DICES-350	210	210×1: 0.272, **105×2: 0.241**, 52×4: 0.242, 26×8: 0.248
DICES-990	594	**594×1: 0.203**, 297×2: 0.205, 148×4: 0.207, 74×8: 0.213, 40×15: 0.219
Dorn Java^{SE}	72	72×1: 0.163, **36×2: 0.163**
EmoBank	1800	**1800×1: 0.088**, 900×2: 0.089, 450×4: 0.090
HelpSteer2	2870	**2870×1: 0.201**, 1436×2: 0.202, 718×4: 0.204
POPQUORN off.	819	**819×1: 0.133**, 410×2: 0.136, 273×3: 0.137, 136×6: 0.140
SummEval coh.	960	960×1: 0.172, **480×2: 0.170**, 320×3: 0.170
ChaosNLI MNLI	959	**959×1: 0.226**, 480×2: 0.234, 240×4: 0.260, 120×8: 0.274, 60×16: 0.298, 30×32: 0.323
ChaosNLI SNLI	908	**908×1: 0.244**, 454×2: 0.258, 227×4: 0.298, 114×8: 0.353, 57×16: 0.408, 28×32: 0.445

$\lambda = X/M$ beats the raw histogram (-0.0020 [-0.0022, -0.0019]) but is indistinguishable from the location-only comparator (inverse-variance pooled -0.000019 [-0.000039, $+0.000000$]; descriptive, since the panels share documents and annotators) on every panel except SummEval and EmoBank, where $\hat{\lambda} \approx 0.4$ (on SummEval a consequence of the duplicated crowd annotations, §1) (results/shrinkage/SUMMARY.md, corrected cluster bootstrap). For forecasting a new rater, what a few-rater histogram contributes beyond its location is small.

6 Spending the annotation budget

A natural heuristic is that, if little of the shape is learnable and the location is what the judge learns, every rating after the first on the same item buys mainly a less noisy location, while a rating on a new item buys a new point for the regression the judge is implicitly fitting. We tested the budget question directly, as an observation in its own right, by holding the total number of ratings fixed and trading items for raters.

Frozen-feature heads (eight ordinal and two nominal panels, thirty configurations each; Table 4; CSC is excluded because its 32 conversation contexts leave fewer than the design's minimum of 20 training groups). On four of the five panels with more than 900 training items (all but SummEval), more items with fewer ratings each wins monotonically: on DICES-990 at a budget of 594 ratings, 594 items × 1 rating gives RPS 0.203, 297 × 2 0.205, 148 × 4 0.207, 74 × 8 0.213 and 40 × 15 0.219; POPQUORN, HelpSteer2 and EmoBank follow the same order, SummEval is flat. The direction reverses only on small item sets (Buse–Weimer with 60 training

Table 5: End-to-end LoRA judges (distributional objective; Qwen2.5-7B and, for SummEval, OLMo-2-7B) at fixed rating budgets: test RPS minus the full-data judge, 95% crossed bootstrap over optimizer seeds and test groups. Prospective blocks use independent rating assignments; contrasts are within one block only (LoRA training is not bit-reproducible). On SummEval the held-out evaluator is an expert and the crowd ratings are nearly uncorrelated with the experts, so its rows measure expert dilution rather than rating budget (§5).

panel	design (items × ratings)	ratings	RPS	− full data
POPQUORN, prospective (Qwen2.5, R20-4)	819 × 1	819	0.1278	+0.0063 [+0.0030, +0.0099]
POPQUORN, prospective (Qwen2.5, R20-4)	117 × 7	819	0.1305	+0.0090 [+0.0056, +0.0124]
POPQUORN, prospective (Qwen2.5, R20-4)	819 × 2	1638	0.1247	+0.0032 [+0.0016, +0.0048]
POPQUORN, prospective (Qwen2.5, R20-4)	819 × 7	5733	0.1215	–
SummEval, prospective, expert dilution (Qwen2.5, R19-9)	960 × 1	960	0.1782	+0.0064 [+0.0045, +0.0085]
SummEval, prospective, expert dilution (Qwen2.5, R19-9)	144 × 7	1008	0.1884	+0.0166 [+0.0118, +0.0216]
SummEval, prospective, expert dilution (Qwen2.5, R19-9)	960 × 2	1920	0.1677	−0.0040 [−0.0078, −0.0003]
SummEval, prospective, expert dilution (Qwen2.5, R19-9)	960 × 7	6720	0.1718	–
SummEval, prospective, expert dilution (OLMo-2, R20-3)	960 × 1	960	0.1862	+0.0066 [+0.0044, +0.0089]
SummEval, prospective, expert dilution (OLMo-2, R20-3)	144 × 7	1008	0.1922	+0.0126 [+0.0058, +0.0197]
SummEval, prospective, expert dilution (OLMo-2, R20-3)	960 × 7	6720	0.1796	–
SummEval, single-batch rerun, expert dilution (Qwen2.5, R20-5b/c)	960 × 1	960	0.1777	+0.0064 [+0.0045, +0.0086]
SummEval, single-batch rerun, expert dilution (Qwen2.5, R20-5b/c)	960 × 2	1920	0.1679	−0.0034 [−0.0066, −0.0004]
SummEval, single-batch rerun, expert dilution (Qwen2.5, R20-5b/c)	960 × 7	6720	0.1713	–
SummEval, single-batch rerun, expert dilution (OLMo-2, R20-6)	960 × 1	960	0.1861	+0.0063 [+0.0042, +0.0085]
SummEval, single-batch rerun, expert dilution (OLMo-2, R20-6)	960 × 2	1920	0.1775	−0.0022 [−0.0050, −0.0001]
SummEval, single-batch rerun, expert dilution (OLMo-2, R20-6)	960 × 7	6720	0.1798	–
SummEval, shared assignment (Qwen2.5)	960 × 1	960	0.1664	−0.0027 [−0.0061, +0.0014]
SummEval, shared assignment (Qwen2.5)	480 × 2	960	0.1651	−0.0040 [−0.0084, −0.0000]
SummEval, shared assignment (Qwen2.5)	144 × 7	1008	0.1897	+0.0206 [+0.0163, +0.0247]
SummEval, shared assignment (Qwen2.5)	960 × 7	6720	0.1691	–
POPQUORN, shared assignment (Qwen2.5)	819 × 1	819	0.1300	+0.0087 [+0.0054, +0.0122]
POPQUORN, shared assignment (Qwen2.5)	117 × 7	819	0.1322	+0.0109 [+0.0070, +0.0151]
POPQUORN, shared assignment (Qwen2.5)	819 × 2	1638	0.1246	+0.0033 [+0.0011, +0.0053]
POPQUORN, shared assignment (Qwen2.5)	819 × 7	5733	0.1213	–
SummEval, shared assignment (OLMo-2)	960 × 1	960	0.1801	+0.0007 [−0.0056, +0.0069]
SummEval, shared assignment (OLMo-2)	480 × 2	960	0.1778	−0.0016 [−0.0062, +0.0023]
SummEval, shared assignment (OLMo-2)	144 × 7	1008	0.1896	+0.0102 [+0.0049, +0.0156]
SummEval, shared assignment (OLMo-2)	960 × 7	6720	0.1794	–

items, DICES-350 with 210), where a single rating per item is too noisy a target. The same items-first order holds on ChaosNLI (959 × 1: KL 0.226; 30 × 32: 0.323) — a result that falsified a prediction we had registered, namely that multi-rater items would pay off at fixed budget where the shape is text-predictable. They do not; where tasks differ is the *marginal* value of a second rating on already-covered items (ChaosNLI −15 %, ordinal heads −1 to −3 %), which we report as an observation.

End-to-end LoRA judges (Table 5). *POPQUORN, the clean budget case.* On POPQUORN offensiveness, whose rating positions are homogeneous, a block re-run in one batch with three optimizer seeds × three folds (SHRINKAGE_PROTOCOL Amendment 7; 205 optimizer steps per arm) gives: extra ratings per item help — 819 × 1 is +0.0087 [+0.0054, +0.0122] worse than the full 5,733-rating data and 819 × 2 recovers 62 % of that gap (819 × 2 − 819 × 1 = −0.0054 [−0.0084, −0.0027]) — and at 819 ratings, 819 × 1 and 117 × 7 cannot be distinguished (117 × 7 − 819 × 1 = +0.0022 [−0.0029, +0.0072]); of the three prespecified readings only "a second rating helps" holds. (An earlier two-seed block with a cross-batch reference had suggested an items-first ordering here as well; it is superseded.) A prospective POPQUORN block with independent assignments (ROUND20 R20-4) reproduces all three readings: 819 × 1 − full = +0.0063 [+0.0030, +0.0099]; a second rating recovers 49 % of that gap (819 × 2 − 819 × 1 = −0.0031 [−0.0062, −0.0002]); and 117 × 7 − 819 × 1 = +0.0027 [−0.0024, +0.0074]. Two ratings per item still cost +0.0032 [+0.0016, +0.0048] on POPQUORN. *How to read the SummEval blocks.* Each SummEval item carries three expert ratings and five crowd ratings. The crowd mean is essentially uncorrelated with the expert mean (r = 0.04; Fabbri et al., 2021), and the evaluator is always a held-out expert. A judge trained on the five crowd ratings alone predicts an almost constant score, +0.014 to +0.025 RPS worse than the full-data judge on every fold. A judge trained on the

two other experts' ratings alone is 0.042 better than the full-data judge for expert 1 but no better for expert 2 (ROUND20 R20-7, descriptive; these two arms ran in a separate batch from the reference arms, and their effects of 0.014–0.042 are an order of magnitude above the ≤ 0.0012 rerun variation). On SummEval, therefore, "fewer ratings per item" changes how far two expert ratings are diluted by five crowd ratings that carry almost no expert-aligned signal. The SummEval contrasts below are contrasts between such dilutions, not evidence about rating budgets within one rater population; POPQUORN, whose rating positions are homogeneous, carries that question. On SummEval coherence, with the Chen et al. objective and the three held-out-expert folds of §3, re-run in one batch with three optimizer seeds × three folds (SHRINKAGE_PROTOCOL Amendment 6; results/lora_budget_v2/), every arm trained for the same 240 optimizer steps (effective batch 8, same schedule, no early stopping; on POPQUORN 205), so arms differ only in the number of distinct items and their repetition; budgets are matched up to rounding ($144 \times 7 = 1{,}008$ ratings, which favours the many-rater arm), a judge trained on 960 items × 1 randomly chosen rating (960 ratings) is not distinguishable from the judge trained on all $960 \times 7 = 6{,}720$ ratings (-0.0027 [-0.0061, $+0.0014$], crossed bootstrap over seeds and test documents; in this block the three seeds share one rating assignment per fold, so the interval covers optimiser but not assignment variation — see the prospective block below); 480×2 even appeared better than the full data (-0.0040 [-0.0084, -0.000001], an interval excluding 0 at a seventh of the budget, which we read as a sign of how much a shared assignment can mislead; vs 960×1: -0.0013 [-0.0073, $+0.0039$]), while 144×7 is worse than 960×1 by $+0.0233$ [$+0.0176$, $+0.0285$] — both prespecified readings hold as worded in the protocol ("one rating per item matches the full data", operationalised as an interval that contains 0 or lies below it, i.e. without an equivalence margin; "items first"). An earlier two-seed block whose full-data reference came from a different batch had suggested that 960×1 *beats* the full data (-0.0061); the same-batch re-run does not support that stronger claim. (The earlier block also gave 240×4: $+0.0162$ and 960×2: -0.0064 vs full; these designs were not re-run.) **Prospective block (ROUND19 R19-9).** We declared a margin of $\delta = 0.005$ RPS and three readings before running new seeds (2003–2005), with every seed × fold cell drawing its own items and ratings. Results: $960 \times 1 -$ full $= +0.0064$ [$+0.0045$, $+0.0085$] (90 %: [$+0.0048$, $+0.0082$]), so equivalence (TOST) and non-inferiority both fail; $144 \times 7 - 960 \times 1 = +0.0102$ [$+0.0052$, $+0.0157$], so items-first holds. The single-rating penalty is concentrated in three of the nine cells ($+0.014$ to $+0.018$; the other six give -0.0009 to $+0.0043$). One of the three is a near-collapsed run whose predicted means barely vary (SD 0.07 against 0.23–0.38 elsewhere). Single-rating training is therefore noticeably less stable across assignments, and the shared-assignment null above reflects a favourable draw. Excluding the collapsed run, the mean penalty is still $+0.0052$ (descriptive). An OLMo-2 replication on the same assignments (ROUND20 R20-3) gives $+0.0066$ [$+0.0044$, $+0.0089$], positive in all nine cells ($+0.0003$ to $+0.0121$), so the penalty does not rest on one failed run. Its largest cells overlap with Qwen2.5's in two of three (per-cell r = 0.56). On OLMo-2, $144 \times 7 - 960 \times 1 = +0.0060$ [-0.0007, $+0.0128$]: items-first is not supported on that base in the prospective block. The contrasts depend on which held-out expert scores them (descriptive, ROUND21 R21-6). On Qwen2.5 the single-rating penalty is ≈ 0 when expert 0 evaluates ($+0.0002$) and $+0.007$ / $+0.012$ for experts 1 and 2. On OLMo-2 it is positive for all three ($+0.004$ to $+0.010$). The experts differ in location (mean ratings 2.81, 2.39, 2.05 against a crowd mean of 2.81), so 'the full data' is a crowd-dominated target scored against one expert at a time, and the budget conclusions on SummEval hold conditional on that expert/crowd mixture. Across the three prospective blocks, one rating per item costs 0.0063–0.0066 RPS against the full data. These are two independent assignment sets: SummEval's is shared by the two bases, and POPQUORN's is separate. In an earlier shared-assignment block with OLMo-2-7B on SummEval, 960×1 was not distinguishable from the full data ($+0.0007$ [-0.0056, $+0.0069$]) and 144×7 was worse than 960×1 by $+0.0095$ [$+0.0022$, $+0.0160$] (results/lora_budget_olmo/summary_vs_a1_F1.0.md); the prospective OLMo-2 block is reported above; in a prospective single-batch block on SummEval/Qwen2.5 (ROUND20 R20-5b/c; full, 960×1 and 960×2 trained together), a second rating per item helps ($960 \times 2 - 960 \times 1 = -0.0098$ [-0.0132, -0.0066]; on OLMo-2, in its own single batch (R20-6), -0.0086 [-0.0117, -0.0056]), and 960×2 is not worse than the full data (OLMo-2: -0.0022 [-0.0050, -0.0001]). Its interval even lies below 0 (-0.0034 [-0.0066, -0.0004]), which we do not read as two ratings being better than seven. The effect is evaluator-specific: per held-out expert, $960 \times 2 - 960 \times 1$ is $+0.003$, -0.009 and -0.023, and $960 \times 2 -$ full is $+0.002$, -0.002 and -0.010 (OLMo-2: $+0.003$, -0.002, -0.007). The advantage over the full data comes entirely from expert 2, the expert whose ratings sit lowest relative to the crowd (descriptive). LoRA training on our hardware is not bit-reproducible (per-cell RPS varies by ≤ 0.0012 between identical runs), so all budget contrasts are taken within one batch (the descriptive R20-7 arms excepted). Single-rating training behaves like stochastic label sampling (Gheibi and Ghazizadeh, 2026) and is consistent with the many-once-over-few-many results for hard-label classifiers (Khetan et al., 2018; Lin

et al., 2014); our contribution is the demonstration with distributional targets and end-to-end LLM judges. We do not derive the budget result from §§2–4: a registered mechanism-based prediction failed on ChaosNLI (§1).

Guidance (conditional on task and learner). For training a judge on an ordinal rating task, many ratings for few items were the better buy only on small item sets: on the ordinal panels with more than 900 items frozen-feature heads preferred more items in every comparison except SummEval, where the orderings are flat (within 0.002), and HelpSteer2 on OLMo-2, where 1436 × 2 edges out 2870 × 1 by 0.0007, while on the panels with ≤ 210 training items (Buse–Weimer, DICES-350) the direction reverses; For LoRA judges the clean case is POPQUORN, whose rating positions are homogeneous: a single rating per item is not a free substitute for the full data (+0.006 RPS in a prospective block), a second rating recovers about half of that, and covering items first was not significantly better than seven ratings on a seventh of them (+0.0027 [−0.0024, +0.0074]). Two ratings per item is therefore the safer default than one. The SummEval LoRA contrasts (§5) mix expert and crowd ratings and are not used for this guidance. Where the rater pool mixes populations, as SummEval's experts and crowd do, who rates matters more than how many ratings there are: crowd-only targets carry almost no information about the experts. Raters belong where a reliable reference distribution is itself the deliverable — evaluation sets, and tasks such as NLI where the shape is in the text.

7 Software-engineering panels

Readability, summary quality and code-smell severity are rated on the same few-rater scales, and the same questions arise for code judges. Dorn's readability study (360 snippets in Java, Python and CUDA; 161–267 developers per snippet) gives the densest ordinal panel we know of; its reliability curve matches Buse–Weimer and DICES: the shape beyond mode and mean reaches a split-half reliability of only 0.2–0.3 even with 40 developers per snippet (Figure 1). The LLM-judge validation benchmarks of the field have three raters per item — Crupi et al.'s code summaries, MLCQ's smell severities, Roy et al.'s summaries — and with three raters the split-half reliability of the shape cannot be estimated. A split leaves two ratings per item, and two ratings are fully determined by their exact mode and mean: the mode is z = z_label (among tied modes the one nearest the median, then the lowest) and the mean is the rating sum, so the donor-replacement support is 0 %. The three-rating histograms themselves do carry shape. On these 4-, 5- and 11-level scales, 15–75 % of items share their exact cell with a different histogram (results/support_three.json), but whether that shape is reproducible cannot be measured from three raters. Scalabrino's nine-rater panel sits in between (Table 1). Decoding from frozen features recovers at most a small fraction of it: the own-shape fraction $G_{\text{own}}/X_{\text{own}}$ is 0.10 (upper bound 0.18) on Dorn's readability panel (360 snippets; normalised forecasting gain G/X +0.03 [−0.03, +0.05]), with a small detectable valid-forecast gain (permutation-corrected projected gain > 0, as on HelpSteer2 and EmoBank, where it reflects within-cell location denoising rather than shape), negative on Scalabrino (G < 0, a small over-fitting penalty on 124 snippets; its X interval covers 0, so no ratio is reported), and uninformative on Buse–Weimer (+0.07 [−0.07, +0.26], 100 snippets) and JetBrains naming (+0.01 [−0.37, +0.03], 120 identifiers) (results/decoding_v2/). For SE human studies the practical reading is that with three raters per artifact the reproducibility of the shape cannot be assessed at all, that with 4–9 it is low (Table 1), and that adding raters to a few artifacts does not make the shape of developer disagreement reproducible on these readability and quality scales.

8 Limitations

Base models: the frozen-feature decoding G/X, target-construction and fixed-budget contrasts on the six three-family panels replicate on Qwen3-8B and on the non-Qwen OLMo-2-7B (the own-shape decoding of §2.2 is computed on all three families for SummEval, MNLI and DICES-350 and agrees in reading; the remaining panels and the valid-forecast gain are Qwen2.5 only) (Table 3: ordinal G/X intervals include 0 on all six three-family panels with upper bounds ≤ 10.2 %, ChaosNLI MNLI G/X 0.17 / 0.18 / 0.18 in Qwen2.5 / Qwen3 / OLMo-2; budget orderings agree except HelpSteer2 on OLMo-2, where 1436 × 2 edges out 2870 × 1 by 0.0007; target-construction contrasts agree in sign except one-hot, whose cost differs by up to 0.009 across families), and the end-to-end budget result replicates with an OLMo-2-7B LoRA judge on SummEval (the audit's LoRA blocks remain Qwen2.5-only); frozen-feature heads for breadth, end-to-end judges on two datasets; decoders — we re-tested with per-layer features and greedy layer selection and with one non-linear probe (random-Fourier-feature kernel ridge) on two panels only (SummEval, POPQUORN; R = 5, B = 100, reduced after a smoke run),

not with fine-tuned probes; SummEval's released crowd annotations (identical across the five crowd positions on 807 of 1,600 items, undocumented and not verified upstream; crowd mean r = 0.04 with the expert mean), which make its full-panel reproducibility and decoding numbers artefactual and its LoRA budget contrasts dilution contrasts; POPQUORN's position folds (a user-held-out check with real user ids, evaluators never in any target, confirms the four conclusions on offensiveness and politeness: mode+mean-preserving replacement −0.00004 / +0.00024 (the latter three SEMs above zero, but two orders of magnitude below the mode-only cost), mode-only +0.0022 / +0.0029, one-hot +0.0042 / +0.0050, 4 ratings on a quarter of the items +0.0042 / +0.0030 worse than 1 rating on all); one registered prediction falsified and reported; effect sizes in the third decimal of RPS, to be read against the mode-only penalty of +0.012 and the budget effects of +0.02. Plant: one construction (item log-dispersion linear in one feature direction, on the M2 model), so a detected plant calibrates power for that construction only; at $a = 20$ (Dorn) the matched plant is detected in 1 of 5 draws, at $a = 40$ (DICES-350) in 5 of 5, and there are no plants on NLI. Own-shape at the exact cell: at 20 raters the exact (mode, mean) cell pools items of different latent location, so a text-decodable own-shape can be location-driven (Dorn, §1); the ordinal upper bounds are therefore upper bounds on shape beyond location, and the NLI values may also contain such a component. The X_own null bias on panels with a finite crossed rater pool (about $-1{\cdot}10^{-4}$ on SummEval, diagnosed by simulation in ROUND21 R21-2) makes the uncorrected X_{own} slightly anti-conservative for "no detectable own-shape" (the corrected value is in §2.2); with the larger corrected denominator the recovery fractions would be smaller, so the reported ratios are not flattered by it. Budget on SummEval: the held-out evaluator is one expert at a time, and the contrasts differ by expert (§5), so they hold for this expert/crowd mixture; LoRA training is not bit-reproducible on our hardware (per-cell RPS varies by ≤ 0.0012 between identical runs), and cross-batch contrasts are used only for the descriptive R20-7 arms, whose effects exceed that variation by an order of magnitude. We make no claim that disagreement is noise, that text contains no residual information, that a theoretical rater count exists, or that label structure causes the ordinal–nominal difference.

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